

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -1(Adv-2022-P1-Model)** **Date: 21-12-2025**

Time: 09.00Am to 12.00Pm **GTA-08** **Max. Marks: 180**

21-12-2025_Sr. XL-500, Sr.S60(N&S) & LIIT_Jee-Adv(2022-P1)_GTA-08_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2022-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Matching Type	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Matching Type	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Matching Type	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

1. Let $\alpha,\beta,\gamma,\delta$ be any values in the closed interval $[0,\pi]$ and if the maximum value of the expression

$$\left(1+\sin^2\alpha\right)\left(1+\cos^2\alpha\right)\left(1+\sin^2\beta\right)\left(1+\cos^2\beta\right)\left(1+\sin^2\gamma\right)\left(1+\cos^2\gamma\right)\left(1+\sin^2\delta\right)\left(1+\cos^2\delta\right)$$

is m which is attained for n number of quadruples $(\alpha,\beta,\gamma,\delta)$ then the value of $\left[\sqrt[4]{mn}\right]$ is equal to _____ $([.]G.I.F)$

2. Let $y=y(x)$ be the solution of the differential equation given by

$\left(3xy^2+x^3\right)dy-\left(4y^3+2yx^2\right)dx=0$. Let $y(1)=1$ and if for some $n\in N$, $y(3)\in[n,n+1)$ then the value of n equals_____

3. Let f be a differentiable real valued function on $\mathbb{R}$ such that

$e^{-x}f(x)=\frac{3}{e^2}+4.e^{-x}\int_2^x\sqrt{2t^2+6t+5}dt\ \forall x\in R$ and let $g(x)=f^{-1}(x)$ then the value of $\left[\left|g''(3)\right|\right]$ is equal to _____([.] is G.I.F)

4. If x,y,z are real and if $\frac{x}{y}+\frac{y}{z}+\frac{z}{x}=13-3\sqrt{2}$ and $\frac{x}{z}+\frac{z}{y}+\frac{y}{x}=13+3\sqrt{2}$ then the value of the expression $\left(\frac{x}{2y}+\frac{y}{2z}\right)\left(\frac{y}{3z}+\frac{z}{3x}\right)\left(\frac{z}{5x}+\frac{x}{5y}\right)$ is equal to _____.

5. If $\beta=\lim\limits_{x\to0}\dfrac{1-\frac{1}{2}x^2-\cos\left(\frac{x}{1-x^2}\right)}{x^2-x^2\cos x}$ then the value of sum of digits in 12β equals ____

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6. Let $A=\begin{pmatrix}-2\beta & 0 & 1\\ 3 & 1 & -3\\ 4 & 1 & -3\end{pmatrix}$ where β is a real number. If $A^{2026}+2(\beta+1)A^{2025}+4\beta A^{2024}$ is a singular matrix then the value of $\left|\left[20\beta\right]\right|$ is equal to _____. $([.]G.I.F)$

7. Given ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$. Let $S=0, S'=0$ be two circles each of which touches $y-$axis at origin and also tangent to the ellipse with least radius. The centres of circles are P and Q and $R(x_1,y_1), S(x_2,y_2)$ are points of contact of circles with ellipse $(\text{where } y_1>0, y_2>0)$. If the area of quadrilateral with vertices P,Q,R,S is $\frac{14\sqrt{2}}{\lambda}$, then the value of $3\left[\frac{\lambda}{2}\right]$ $([.]GIF)$ is _____

8. Consider the function $f(x)=\begin{cases}\sin\left(\frac{\pi}{4}(x-[x])\right) & \text{if } [x] \text{is odd}, \quad x\geq 0\\ \cos\left(\frac{\pi}{4}(1-x+[x])\right) & \text{if } [x] \text{is even}, \quad x\geq 0\end{cases}$ $([.]\text{is } G.I.F)$

In the interval $x\in[0,4\pi]$ the number of solutions of the equation $2f(x)-1=0$ is _____.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

9. In $\Delta ABC, BC=8$ and $2\sin A+3\cos B\cos C=4$. Then which of the following is/are correct

A) $B=C$ **B)** $\sin A=\frac{3}{5}$

C) Area of triangle ABC is 1 **D)** $\sin A=\frac{4}{5}$

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10. A 2×3 rectangular plate has vertices $(0,0),(2,0),(0,3)$ *and* $(2,3)$. It rotates 90° clockwise about the point $(2,0)$, it then rotates 90° clockwise about the point $(5,0)$. Then 90° clockwise about $(7,0)$ and finally 90° clockwise about $(10,0)$. Let S be the curve traced by the point on the plate which is initially located at $(1,1)$. The area bounded by S, x-axis between lines $x=1 \& x=11$ is $a+\frac{b}{2}\pi$ then $(a,b\in N)$ Which is/are correct

A) $a=6$ **B)** $b=7$ **C)** $a=5$ **D)** $b=9$

11. Let there be 3 lines L_1,L_2,L_3 given by

$L_1: x+2y-z-2=0=2x-y+3z-4$

$L_2: 2x+y-z-2=0=x-4y+\lambda z-1 \quad (\lambda \text{ is real})$

$L_3: x=0=y$

If the shortest distance between L_i *and* L_j $(i\neq j \,\&\, i,j=1,2,3)$ id denoted by $d(L_i,L_j)$ then (L_1,L_2 are coplanar lines). Which is/are correct ([.] GIF)

A) $\left[d(L_1,L_3)\right]=1$ **B)** $\left[d(L_1,L_3)\right]=0$

C) $d(L_1,L_3)>d(L_2,L_3)$ **D)** $d(L_1,L_3)<d(L_2,L_3)$

12. Let $A=(a_{ij})$ be a 3×3 matrix such that

$$a_{ij}=\begin{cases}\underset{x\to0}{Lt}\dfrac{\left(1+3x+2x^2\right)^{1/x}-\left(1+3x-2x^2\right)^{1/x}-4(1+3x)^{1/x}+x}{x} & ,i=j\\ \sin^{25}\left(\left(i^{22}-j^{22}\right)x\right)+2\tan^{-1}\left(e^{(i-j)x}\right)-\dfrac{\pi}{2} & ,i\neq j\end{cases}$$. Then which is/are correct

A) $\det(A-I)=0$ **B)** $\det(A+I)=0$

C) $\det\left(A^2+A-2I\right)=0$ **D)** A is skew-symmetric matrix

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13. If $J=\int_0^{\pi/4}\frac{e^{\sec x}\sin\left(x+\frac{\pi}{4}\right)}{(1-\sin x)\cos x}dx$ and if the value of J equals $\frac{\left(a+\sqrt{b}\right)e^{\sqrt{c}}-e}{\sqrt{d}}$ (where e is exp) then which is/are correct $(a,b,c,d\in N)$

A) $a+b=3$ **B)** $c+d=4$ **C)** a is not prime **D)** c is odd

14. Let $A=\{1,2,3,4,5,6,7\}, B=\{1,2,3,4,5,6\}$ let $f:A\to B$ be a function then which is/are correct

A) Probability that f is a surjection is $\frac{35}{648}$

B) Probability that f is injection is 0.

C) Probability that the $n(f(A))=2$ is $\frac{35}{5184}$

D) Probability that f is a non-decreasing function is $\frac{1}{2}$

SECTION – III
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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15. Match the Column – I & II

Column – I		Column – II	
(A)	If $\sqrt{3}, H, \sqrt{2}$ are in H.P, then the value of $3\left(\frac{H+\sqrt{2}}{H-\sqrt{2}}+\frac{H+\sqrt{3}}{H-\sqrt{3}}\right)$ is	**(P)**	6
(B)	The value of $\sum_{n=1}^{\infty}\sum_{k=1}^{\infty}\frac{k}{2^{n+k}}$ is	**(Q)**	$\frac{3}{2}$
(C)	If $L=\lim_{x\to 0}\frac{x\sin(\sin x)-\sin^2 x}{x^6}$ then $72L$ is equal to	**(R)**	4
(D)	Let $f(x)=\begin{cases} x-2n, & 2n\le x<2n+1 \\ \frac{1}{2}, & 2n+1\le x<2n+2 \end{cases}$ then The fundamental period of $f(x)$ is	**(S)**	2
		T)	0

A) A – P, B – Q, C – T, D – R **B)** A – P, B – S, C – R, D – S

C) A – Q, B – R, C – S, D – P **D)** A – Q, B – S, C – R, D – P

16. Let R_1, R_2 be the relations on the set $\{1,2,3,4,5,.....,46\}$ such that

$R_1=\left\{\left(\left(p,p^n\right): p \text{ is prime}, n\ge 0, n \text{ is an integer}\right)\right\}$

$R_2=\left\{\left(\left(p,p^n\right): p \text{ is prime and } n=0 \text{ or } 1\right)\right\}$

Let n_1 and n_2 denote the number of ordered pairs, in relations R_1 and R_2 respectively.

Let n_3 denotes the minimum number of ordered pairs which must be added to the relation $R_3=\{(1,2),(2,3)\}$ on the set $\{1,2,3\}$ so that R_3 is equivalence relation then match the following

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Column – I		Column – II	
(A)	Number of positive integral divisors of n_1	**(P)**	2
(B)	Number of positive integral divisors of n_2	**(Q)**	4
(C)	Number of positive integral divisors of $n_2 + n_3$	**(R)**	6
(D)	Number of positive integral divisors of $n_1 + n_2 + n_3$	**(S)**	9

A) A – S, B – R, C – Q, D – P **B)** A – P, B – Q, C – R, D – S

C) A – S, B – P, C – S, D – R **D)** A – P, B – Q, C – R, D – P

17. A tangent with slope m drawn to the parabola $y^2 = -ax, (a > 0)$ from a point $A(1,0)$ which also touches the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ at B such that $\angle ASB = 90°$ where S is the focus of hyperbola then match the following.

Column – I		Column – II	
(A)	The value of sum of digits of a	**(P)**	2
(B)	The value of sum of digits of b^2	**(Q)**	4
(C)	The value of m^2	**(R)**	3
(D)	$a - b^2$ equals	**(S)**	7

A) A – Q, B – R, C – S, D – P **B)** A – P, B – Q, C – S, D – R

C) A – P, B – S, C – R, D – S **D)** A – S, B – R, C – Q, D – Q

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18. Match the Column – I & II

Column – I		Column – II	
(A)	The least value of $f(x)=\sqrt{x^2-2x+2}+\sqrt{x^2-4x+29}$ occurs at $x=\alpha$ then 6α is	**(P)**	3
(B)	The least value of $f(x)=\sqrt{x^2-2x+2}+\sqrt{x^2-4x+29}$ equals m then $[m]$ $([.]GIF)$	**(Q)**	5
(C)	$A=\{1,2,3,4,5,6,7\}$, $P(A)$ is power set of A then number of $f:A\to P(A)$ such that $a\in f(a),\forall a\in A$ is 2^K $(K\in N)$ so that $K/6$ equals	**(R)**	6
(D)	$x\in[0,5\pi]$, number of solutions of $3\sin^2 x+\cos^2 x+\sqrt{3}\sin x+\cos x+1=\sqrt{3}\sin x\cos x$ is K then $K+3$ equals	**(S)**	7

A) A – P, B – Q, C – S, D – R
B) A – P, B – S, C – R, D – S
C) A – S, B – R, C – S, D – P
D) A – Q, B – S, C – R, D – P

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PHYSICS **Max Marks: 60**

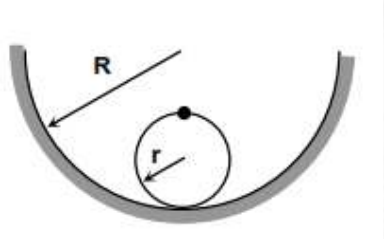
SECTION – I
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions.

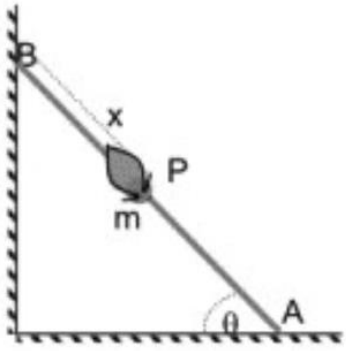
- The answer to each question is a **NUMERICAL VALUE.**
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- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

19. A small object is mounted to the perimeter of a hoop of radius r. The mass of the object and the hoop is the same. The hoop is placed into a fixed semi-cylinder shaped rough trough of radius R, such that the small mass is at the top. Find the maximum value of R/r ratio such that the object performs oscillatory motion for small displacement.

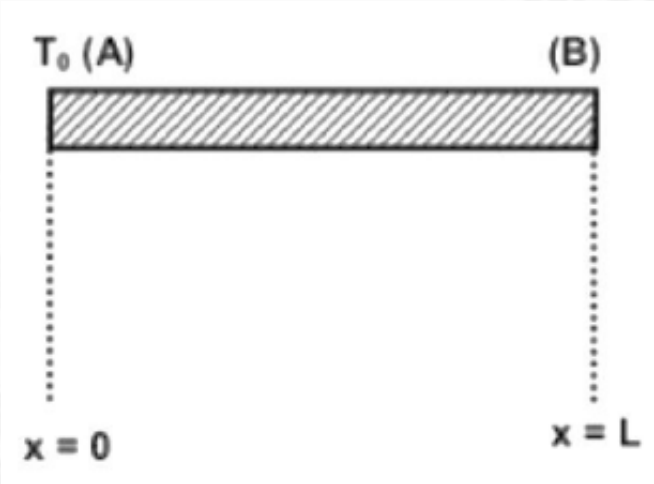


20. A light stick of length l rests with its one end against the smooth wall and other end against the smooth floor and other end against the smooth horizontal floor as shown in the figure. The bug starts at rest from point B and moves such that the stick always remains at rest. The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\pi\sqrt{\frac{l\sin\theta}{Ng}}$. Determine value of N.



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21. Figure shows a metal rod of uniform cross section area A, with variable thermal conductivity given by $k(x) = k_0 \sec\left(\frac{\pi}{6L}x\right)$. If the end A is maintained at temperature T_0, the rod carries a thermal current I_0 (from B to A) in steady state and $\frac{I_0 L}{k_0 A T_0} = \frac{\pi}{3}$; find the temperature of the end B of the rod. Let's say this temperature is kT_0, find integer value k.

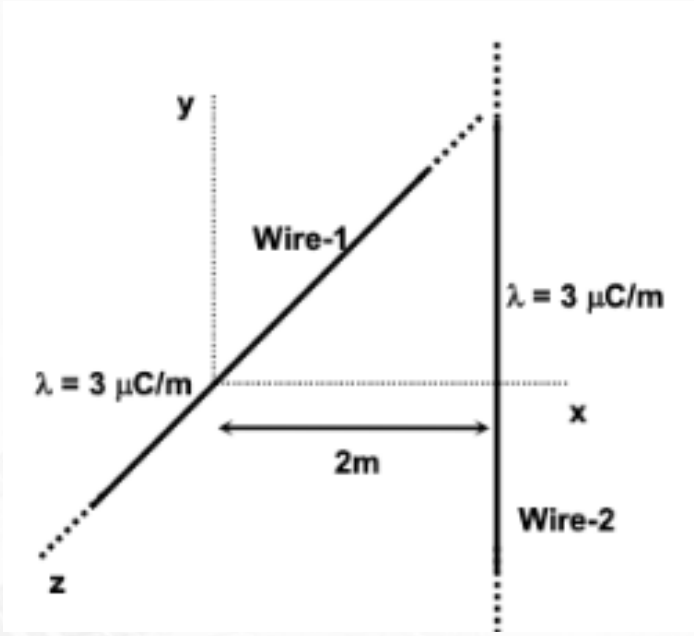


22. A snake of length $4m$ slithered half of its length into a narrow, straight tube. The outside end of the snake can curve arbitrarily on the horizontal ground. The snake is modeled by a flexible rope of length $4m$ with a uniform mass distribution. The area of region where centre of mass of the snake can lie is Nm^2. Determine N.



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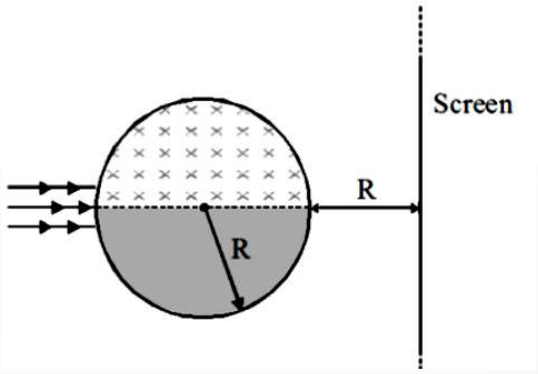
23. A long wire has uniform charge density $\lambda = 3\mu C/m$ is kept along z-axis. Another similar wire 2 is lying on x-y plane such that the minimum separation between them is $d = 2m.$ Calculate the work done (in Joule) in moving the wire-2 upto origin keeping the wire parallel to initial condition.



24. Two wires of same length but of different square cross sections are made from the same material. The sides of the cross sections of the first and second wires are $d_1 = 1mm$ and $d_2 = 4mm.$ The current required to fuse the first wire is $I_1 = 1.0\,A.$ Determine the least current $I_2(\text{in } A)$ required to fuse the second wire, assuming that the amount of heat dissipated to the ambient per second obeys the law $Q = 9kS(T - T_{am}),$ where S is the surface area of the wire, T is its temperature, T_{am} is the temperature of the ambient away from the wire, and k is the proportionality factor which is the same for the two wires.

25. Figure shows circular region of radius $R = \sqrt{3}m$ in which upper half has uniform magnetic field $\vec{B} = 0.2(-\hat{k})T$ and lower half has uniform magnetic field $\vec{B} = 0.2\hat{k}T.$ A very thin parallel beam of point charges each having mass $m = 2gm,$ speed $v = 0.3m/\sec$ and charge $q = \pm 1mC$ are projected along the diameter as shown in figure. A screen is placed perpendicular to initial velocity of charges as shown. If the distance between the points on screen where charges will strikes after being deflected by the magnetic field is $4X$ meters, then calculate $X.$

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26. In the mid-19th century, French physicist Foucault made the most accurate measurement of the speed of light using a laboratory-sized apparatus consisting of rotating and fixed mirrors. Consider a source of light emitting a beam which bounces off a rotating mirror and then reflects off a small fixed mirror. By the time it reaches back to the rotating mirror, the mirror has rotated by a small angle. This rotation will deflect the beam through a small angle 30 minute of an arc from its original path. If the mirror rotates with a frequency $f\left((5/3)\times10^4 Hz\right)$ and the speed of light is given as $3\times10^8\, m/\sec$, the value of D (where D is distance between rotating mirror and fixed mirror) is $'a'meter$ then the value of a is?

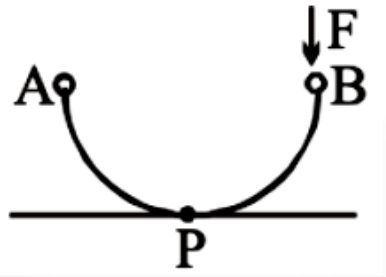
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

27. A half section of pipe of mass m and radius r rests on a rough horizontal surface. A vertical force F is applied as shown.

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Assuming that the section rolls without sliding. Then [Centre of mass of half ring is $\frac{2R}{\pi}$ below the centre]

A) The moment of inertia about an axis passing through P and parallel to the axis of pipe is $\frac{2mr^2}{\pi}(\pi-2)$

B) For the instant shown, the angular acceleration is $\frac{F\pi}{2mr(\pi-2)}$

C) The moment of inertia about an axis passing through A and parallel to the axis of pipe is $2mr^2$

D) Torque due to applied force acts in the clock wise direction.

28. A rod of length ℓ, has linear coefficient of thermal expansion α which is given as $\alpha=\alpha_0+\alpha_1 x+\alpha_2 T$. The rod is heated from temperature T_0 to T_1. Then choose the INCORRECT statement(s).

A) Dimensional formulae of α_0,α_1 & α_2 are different.

B) Change in length of small element of length 'dx' is different for different portion of rod.

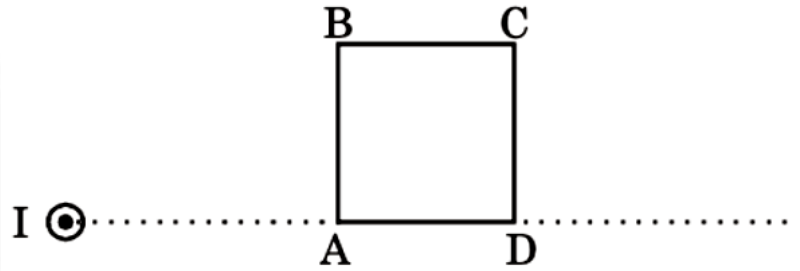
C) Total elongation in the rod varies linearly with length 'ℓ' of the rod.

D) Net elongation in the rod is given by

$$\Delta\ell=\alpha_0\ell(T_1-T_0)+\alpha_1\frac{\ell}{2}(T_1-T_0)+\alpha_2\ell(T_1^2-T_0^2)$$

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29. There is an infinitely long wire carrying current in the direction perpendicular to the plane of paper as shown in the figure. Also shown is a square amperian path $ABCD$. Mark the **CORRECT** options:



A) $\left|\int_{D}^{C}\vec{B}.d\vec{\ell}+\int_{C}^{B}\vec{B}.d\vec{\ell}\right|<\left|\int_{D}^{C}\vec{B}.d\vec{\ell}\right|$

B) $\left|\int_{A}^{B}\vec{B}.d\vec{\ell}\right|>\left|\int_{D}^{C}\vec{B}.d\vec{\ell}\right|$

C) $\left|\int_{A}^{B}\vec{B}.d\vec{\ell}+\int_{B}^{C}\vec{B}.d\vec{\ell}\right|=\left|\int_{A}^{D}\vec{B}.d\vec{\ell}+\int_{D}^{C}\vec{B}.d\vec{\ell}\right|$

D) $\left|\int_{A}^{D}\vec{B}.d\vec{\ell}\right|=0$

30. A nucleus has a radius $7.2\times10^{-15}m$. When an $\alpha-$decay takes place from this nucleus, ratio of number of neutrons and number of protons in the daughter nucleus becomes $\frac{65}{41}$. $\left(R_0=1.2\times10^{-15}m,\text{where}\,R_0\,\text{is the radius of hydrogen nucleus}\right)$

A) Parent nucleus is ${}_{84}Po$.

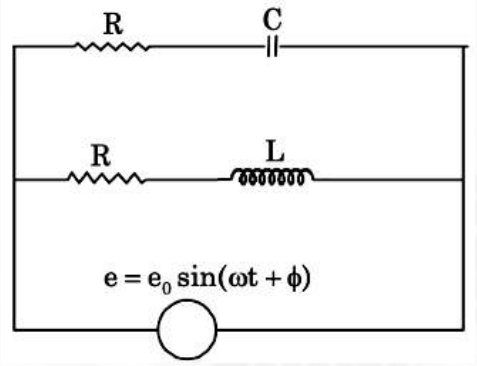
B) Daughter nucleus is ${}_{82}Pb$.

C) Mass number of daughter nucleus is 216

D) Mass number of parent nucleus is 216.

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31. A loud speaker system uses alternating current to amplify sound of certain frequencies. It consists of two speakers. Tweeter – which has smaller diameter produces high frequency sounds. Woofer – which has larger diameter produces low frequency sound. For purpose of circuit analysis, we can take both speakers to be of equal resistance R. The equivalent circuit is shown in the figure. Select the correct alternative.



A) The speaker connected with the capacitor acts as woofer.

B) The speaker connected with the inductor act as tweeter.

C) For the frequency, $\frac{1}{2\pi\sqrt{LC}}$ the sound intensity is same in both tweeter and woofer.

D) When $L = R^2C$, phase difference between currents in the woofer and tweeter is 90°.

32. A man is standing on a plank which is placed on smooth surface. There is sufficient friction between feet of man and plank. Now man starts running over plank.



The correct statement(s) is/are

A) Work done by friction on man with respect to ground is negative

B) Work done by friction on man with respect to ground is positive.

C) Work done by friction on plank with respect to ground is positive.

D) Work done by friction on man with respect to plank is zero.

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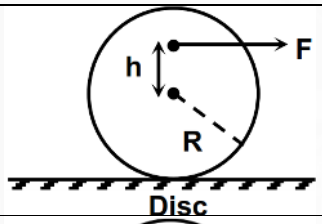
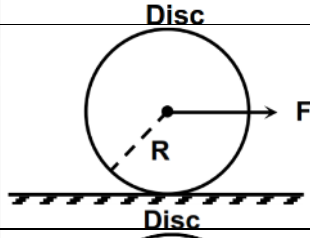
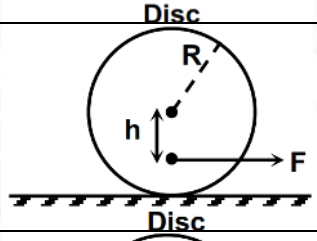
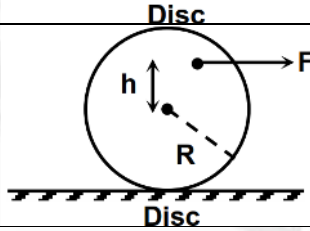
SECTION – III
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

33. In each case, there is sufficient friction for regular rigid uniform body to undergo pure rolling on a rigid horizontal surface. Now Match the Column I and II

Column – I		Column – II	
(A)		**(P)**	The direction of static friction may be forward or may be backward or static friction may be zero
(B)		**(Q)**	The direction of static friction is towards backward
(C)		**(R)**	The angular acceleration will be clockwise
(D)		**(S)**	Acceleration of the centre mass will be along direction 'F'

A) $(A)\to P,R;(B)\to P,S;(C)\to Q,R;(D)\to Q,S$

B) $(A)\to P,S,R;(B)\to Q,R,S;(C)\to Q,R,S,(D)\to P,R,S$

C) $(A)\to Q,R,S;(B)\to P,R,S;(C)\to Q,R,S,(D)\to P,S,R$

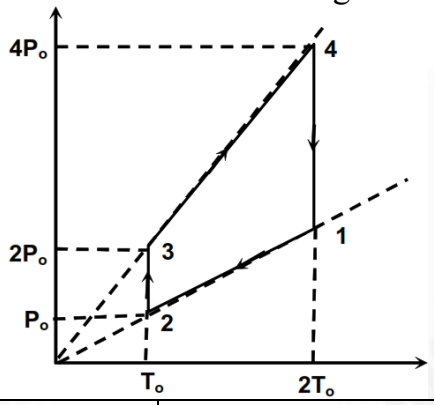
D) $(A)\to Q;(B)\to P,R;(C)\to Q,S;(D)\to P,S$

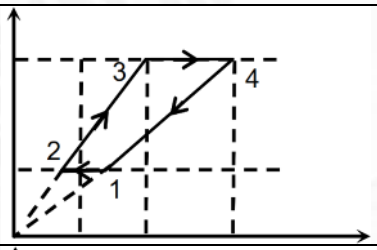
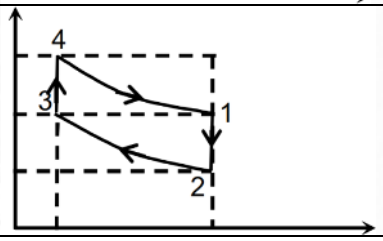
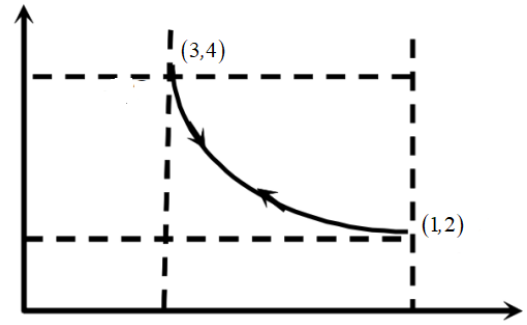
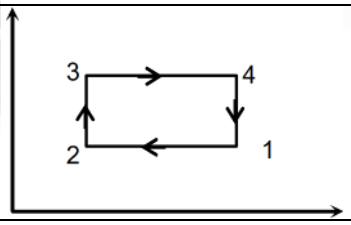
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34. Read the following and write the correct pairs.

Pressure temperature graph of n moles of an ideal gas is shown.



Column – I		Column – II	
(A)	$\rho - V$ graph	**(P)**	
(B)	$P - V$ graph	**(Q)**	
(C)	$\rho - P$ graph	**(R)**	
(D)	$V - T$ graph	**(S)**	

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A) $(A)\to Q,R;(B)\to P;(C)\to S;(D)\to Q$ **B)** $(A)\to Q;(B)\to S;(C)\to P;(D)\to R$

C) $(A)\to R;(B)\to Q;(C)\to P;(D)\to S$ **D)** $(A)\to PR;(B)\to QP;(C)\to Q;(D)\to S$

35. A steel rope (density d, length L, area of cross section A. Young's modulus Y) is subjected to operations as described in column-I. Discover the correct value of x in the expressions and match.

	Column – I		Column – II
(A)	The rope is hanging from a ceiling in equilibrium. The elastic potential energy fper unit volume at a point $L/3$ from the ceiling is $\frac{x}{18}\frac{(dLg)^2}{Y}$.	**(P)**	3
(B)	The rope is vertical and moves under the application of vertical force $F=\frac{dALg}{2}$ acting at its upper end upwards. The strain at a point at a distance $\frac{L}{3}$ from the lower end is $\frac{dLg}{xY}$	**(Q)**	4
(C)	The rope is rotating at angular speed ω_0 in gravity free space about its one fixed end. The stress at a point $\frac{L}{3}$ from the fixed end is $\frac{x}{18}\left(dL^2\omega_0^2\right)$	**(R)**	5
(D)	A circular ring is formed from the rope. The ring is made to rotate with angular speed ω_0 in its own plane in gravity free space. Tension in the rope is $x\left(\frac{dAL^2\omega_0^2}{20\pi^2}\right)$	**(S)**	6
		(T)	None of the these

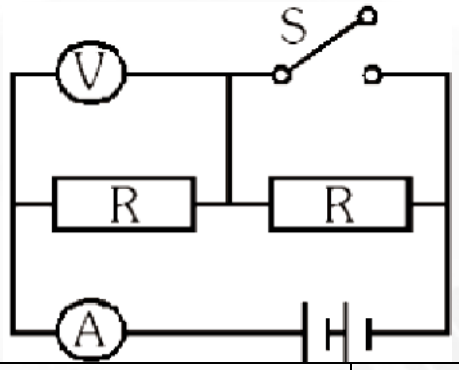
A) $(A)\to Q,(B)\to S;(C)\to T;(D)-R$ **B)** $(A)\to T,(B)\to Q;(C)\to S;(D)-R$

C) $(A)\to Q,(B)\to P;(C)\to R;(D)-T$ **D)** $(A)\to S,(B)\to Q;(C)\to P;(D)-R$

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36. The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed?



	Column – I		Column – II
(A)	Ammeter Reading	**(P)**	Increases
(B)	Voltmeter Reading	**(Q)**	Decreases
(C)	Equivalence resistance of circuit	**(R)**	Doe not change
(D)	Power dissipated across R in right branch	**(S)**	Becomes zero
		(T)	Cannot be determined.

A) $(A)\to P;(B)\to P;(C)\to Q;(D)\to QS$

B) $(A)\to S;(B)\to Q;(C)\to PR;(D)\to P$

C) $(A)\to P;(B)\to Q;(C)\to QS;(D)\to RS$

D) $(A)\to PQ;(B)\to Q;(C)\to S;(D)\to R$

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CHEMISTRY **Max Marks: 60**

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions.

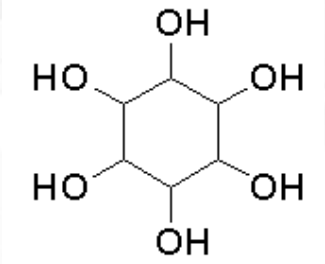
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

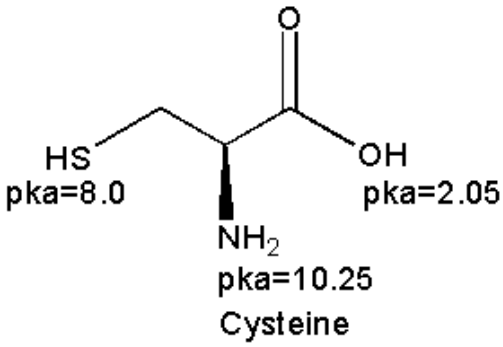
37. Benzaldehyde does not yield a simple addition product with NH_3. 'X' is the stable organic product isolated from the reaction between benzaldehyde and NH_3. In the compound 'X', number of *pi* bonds in conjugation with the imine (>C=N) functional groups (s) is/are:

(Dont include pi bonds of >C=N)

38. How many geometrical isomers are possible for inositol? If the answer is more than one digit number, then answer the sum of digits?



39. At what *pH* cysteine migrates neither towards anode nor cathode?



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40. 2 moles of an ideal gas at 400K undergoes isobaric expansion from 1L to 4L. The ratio of heat supplied (to the gas) to work done by the gas is 7:2. What is the value of molar specific heat capacity at constant volume for the gas $\left(\text{in cal mol}^{-1}\text{K}^{-1}\right)$.

$\left(\text{Take } R = 2\,\text{Cal mol}^{-1}\text{K}^{-1}\right)$

41. A solution is prepared by dissolving $4g\ NaOH$, $4.6g\ HCOOH$ and $6g\ CH_3COOH$ and a few drops of the indicator phenolphthalein in 500 ml water and diluting upto 1L. The percentage of phenolphthalein that exists in the coloured form is $x\times10^{-y}$, where x and y are single digit integers. Calculate $|x-y|$.

$\left(K_a \text{ of } HCOOH \text{ is } 10^{-4}; K_a \text{ of } CH_3COOH \text{ is } 2.5\times10^{-5}; K_a \text{ of phenolphthalein is } 4\times10^{-10}\right)$.

42. Magnetite $\left(Fe_3O_4\right)$ can be converted into metallic iron by heating with carbon monoxide. How many kilograms of Fe_3O_4 must be processed to get 4.2kg iron if the process is 74% efficient?

At wts $Fe=56$ $O=16$ $C=12$

43. Determine the crystal field stabilization energy of d^6 complex having $\Delta = 25000cm^{-1}$ and $P=15000cm^{-1}$. If your answer is $A\times10^B$, What is the value of $A+B$?

44. The spin magnetic moment of $MnSO_4 5H_2O$ is

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

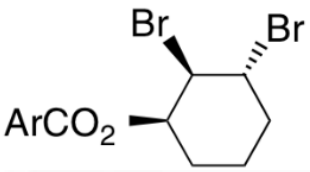
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

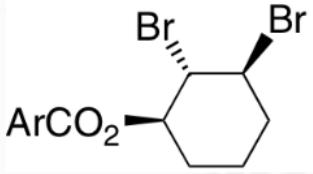
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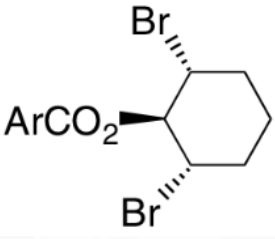

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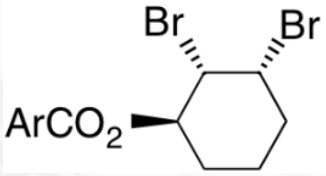
45. The bromination of 3 – aroyloxycyclohexenes gives rise to a mixture of stereoisomeric and regioisomeric products. Which of the following presents in the product mixture? (Ar is phenyl).



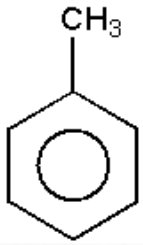
A) 

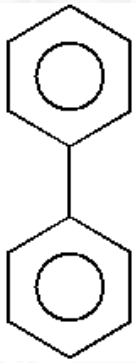
B) 

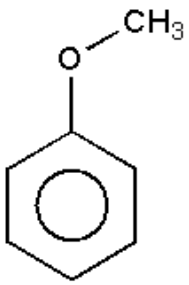
C) 

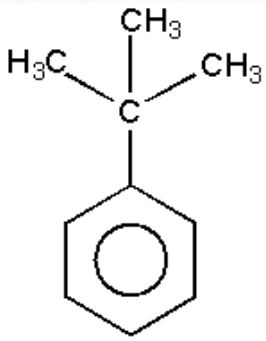
D) 

46. In aromatic electrophilic substitution at meta position which of the following is/are more reactive than benzene?

A) 

B) 

C) 

D) 

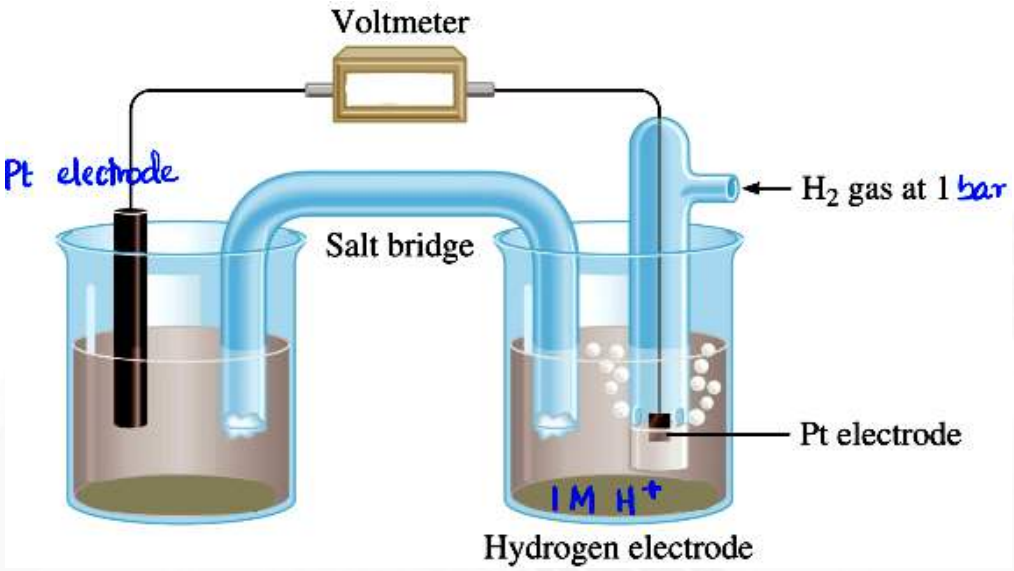
47. To a solution containing $1.1\text{L}10^{-2}\text{M}\,FeSO_4$, 1 ml of 1M $Ce(NO_3)_4$ solution is added. The following reaction occurs.

$$Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$$

This reaction mixture(taken in the beaker on the left) is now coupled with a standard hydrogen electrode as shown
below. Assume that the salt bridge eliminates the liquid junction potential.

$$\left(E^{\circ}_{Ce^{4+},Ce^{3+}} = 1.43V; E^{\circ}_{Fe^{3+},Fe^{2+}} = 0.77V; 2.303\frac{RT}{F} = 0.06\right)$$

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Which of the following statements are correct, if it is known that theoretical calculation of E_{cell} agrees exactly with the voltmeter reading.

A) For the solution on the left $E_{Fe^{3+},Fe^{2+}} = E_{Ce^{4+},Ce^{3+}}$

B) $\left[Ce^{4+}\right]$ in the left container is 10^{-12} M.

C) The voltmeter reading can be calculated by using the equation

$$E = E^{\circ}_{Fe^{3+},Fe^{2+}} - 0.06\log\frac{\left[Fe^{2+}\right]}{\left[Fe^{3+}\right]}$$

D) The voltmeter reading can be calculated by using the equation

$$E = E^{\circ}_{Ce^{4+},Ce^{3+}} - 0.06\log\frac{\left[Ce^{3+}\right]}{\left[Ce^{4+}\right]}$$

48. A solution of 2 liquids A and B follows Raoult's law over the entire range of composition. The vapour pressures of the liquids are P°_A and P°_B respectively. A solution is prepared by mixing equal moles of A and B (solution 1). A different solution containing equal masses of A and B is prepared (solution 2). Any mixture of the solutions 1 and 2 has higher vapour pressure than solution 2. Which of the following **must be correct?**

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A) If $MM_A > MM_B$, then $P_A^\circ > P_B^\circ$ $(MM = \text{molar mass})$

B) If $MM_A > MM_B$, then $Y_{A_1} > Y_{B_1}$ (Y_{A_1} and Y_{B_1} are mole fraction of vapours of A and B in equilibrium above solution 1).

C) If $MM_A > MM_B$, then $Y_{A_2} > Y_{B_2}$ (Y_{A_2} and Y_{B_2} are mole fraction of vapours of A and B in equilibrium above solution 2).

D) If $MM_A > MM_B$, then $P_A^\circ < P_B^\circ$ $(MM = \text{molar mass})$

49. Iron pyrites is a mineral containing iron and sulphur. The correct statement regarding this mineral is/are

A) Metallic ion shows paramagnetic with spin magnetic moment of 4.9BM

B) The sulphur exist as diatomic dianion.

C) Oxidation state of iron ion in the mineral is +2.

D) On burning, it forms FeO and SO_2

50. Which of the following statements is/are correct regarding OCl_2 and ClO_2

A) Both OCl_2 and ClO_2 have same shape,

B) In both OCl_2 and ClO_2, $Cl-O$ bond lengths are equal.

C) In both OCl_2 and ClO_2, the hybridisation of central atom is same.

D) Both OCl_2 and ClO_2, are good oxidizing and bleaching agents.

SECTION – III
(MATCHING TYPE)

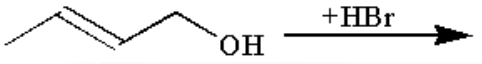
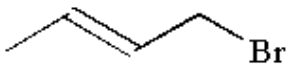
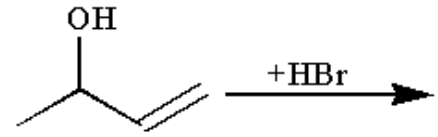
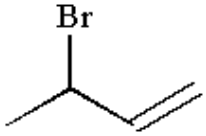
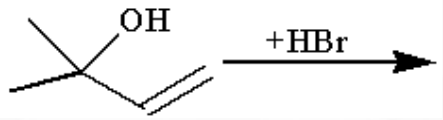
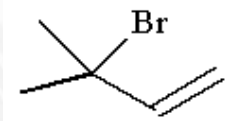
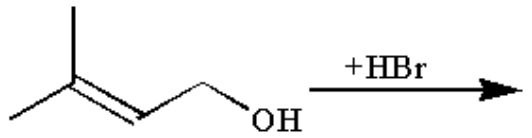
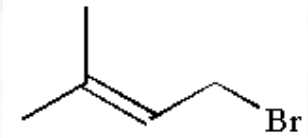
This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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51. In the LIST– I some reactions given, in the LIST – II major product of these reactions given.

LIST – I		LIST – II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	

The correct match is

A) $I \to P; II \to Q; III \to R; IV \to S$ **B)** $I \to P; II \to P; III \to S; IV \to S$

C) $I \to Q; II \to Q; III \to R; IV \to R$ **D)** $I \to Q; II \to P; III \to S; IV \to R$

52. Match the following

Column I		Column II	
(A)	$Al(OH)_3$ *Sol*	(P)	Negatively charged colloidal particle
(B)	As_2S_3 *sol*	(Q)	Positively charged colloidal particles
(C)	Gold Sol	(R)	Lyophillic Sol
(D)	Starch	(S)	Lyophobic Sol

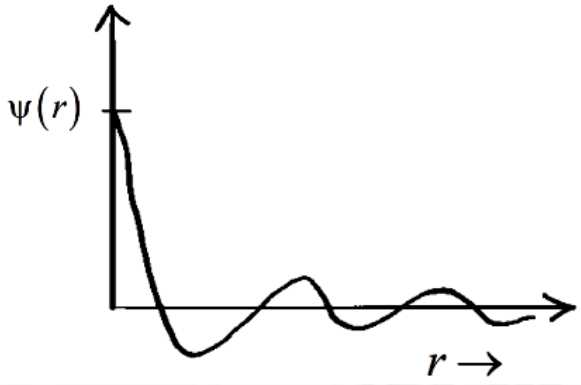
A) A-QS, B-PS, C-PS, D-PR **B)** A-PS, B-QS, C-PS, D-PR

C) A-QS, B-PS, C-PR, D-PS **D)** A-PS, B-QS, C-PR, D-QR

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53. The wave function of an electron in a single electron species X^{n+} is represented as follows (X belongs to either first or second period).



A gaseous sample of X^{n+} containing electrons in only the above excited state is allowed to de-excite upto the ground state. The emitted photons are allowed to fall on a sample of ground state H atoms. This may result in ionization or excitation of the H atom on which the photon falls. Match X^{n+} with the possible states of electrons (principal quantum number)of H sample after the emitted photons fall on the sample.

List – I (X^{n+})		**List – II** (**Possible states of electron**)	
(A)	H	(P)	$n=1$
(B)	He^{+}	(Q)	$n=2$
(C)	Li^{2+}	(R)	$n=3$
(D)	Be^{3+}	(S)	$n=4$
		(T)	Ionization takes place

A) A-PQRS, B-PQRT, C-PQT, D-PT **B)** A-PQRS, B-PQRT, C-PQT, D-PST

C) A-PQRS, B-PQT, C-PQRT, D-PT **D)** A-PT, B-PQT, C-PQT, D-PST

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54. Match the pair of complex compounds, the properitees that are same, then

COLUMN – I		COLUMN – II	
A)	$\left[Mn(H_2O)_4\right]SO_4.H_2O$ & $\left[Mn(CN)_4\right]^{2-}$	**P)**	Hybridisation
B)	$\left[Mn(CO)_4\right]^{3-}$ & $\left[Fe(CO)_4\right]^{2-}$	**Q)**	Sum of magnitude of charges
C)	$\left[Co(C_2O_4)_3\right]^{3-}$ & $\left[Co(H_2O)_6\right]^{3+}$	**R)**	Geometry
D)	$[NiCl_4]^{2-}$ & $[PdCl_4]^{2-}$	**S)**	Magnetic moment
		T)	Electrical conductivity

A) A – PQRST, B – PQRS, C – PRT, D – Q

B) A – PQRS, B – PRS, C – PQRS, D – Q

C) A – PS, B – PRST, C – PQRST, D – Q

D) A – PQRS, B – PRST, C – PQRT, D – Q

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